Bias and Study Limitations

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Common types of bias

Selection bias

The people included in the study differ systematically from the population we want to understand.
A survey about physical activity is advertised only through a university sports club.
→ Participants may be more active than the average student.



Measurement bias

The way something is measured systematically affects the results.
Patients are asked to recall their physical activity from the previous year.
→ Recall may be inaccurate.



Common types of bias

Reporting / response bias

Participants may give answers that are socially desirable or may selectively report information.

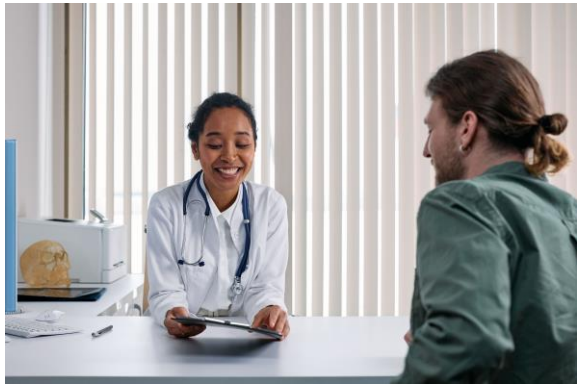
People underreport alcohol consumption.



Observer / researcher bias

The researcher's expectations influence measurement, observation, or interpretation.

An observer knows which patients received the intervention and unconsciously rates their behavior more positively.



Publication bias

Positive results get more frequently published than negative results.

Bias vs limitation

Bias	Limitation
Systematic error	Weakness or constraint
Can distort findings	Can reduce precision/generalizability
Often threatens validity	May affect interpretation
Example: selection bias	Example: small sample

This distinction is important, but they can overlap.

Example: missing data
If 30% of participants don't provide follow-up data, this is a **limitation**.

If the people who drop out are systematically different from those who remain, it can create **attrition bias**.



Other common study limitations

- **Small sample size** → less precise estimates
- **Missing data** → incomplete information
- **Short follow-up** → long-term effects may be unknown
- **Single-center study** → limited generalizability
- **Self-reported data** → recall/reporting problems
- **No control group** → difficult to attribute effects to an intervention
- **Confounding** → another factor may explain the observed association

Group case analysis

Study: Researchers investigate whether a new exercise program reduces blood pressure. They recruit 40 volunteers from a fitness center. Blood pressure is measured only once, immediately after the program. Ten participants drop out. There is no control group.

1. **What are the limitations?**
2. **What potential biases are present?**
3. **How might these affect the findings?**
4. **What could the researchers have done differently?**

